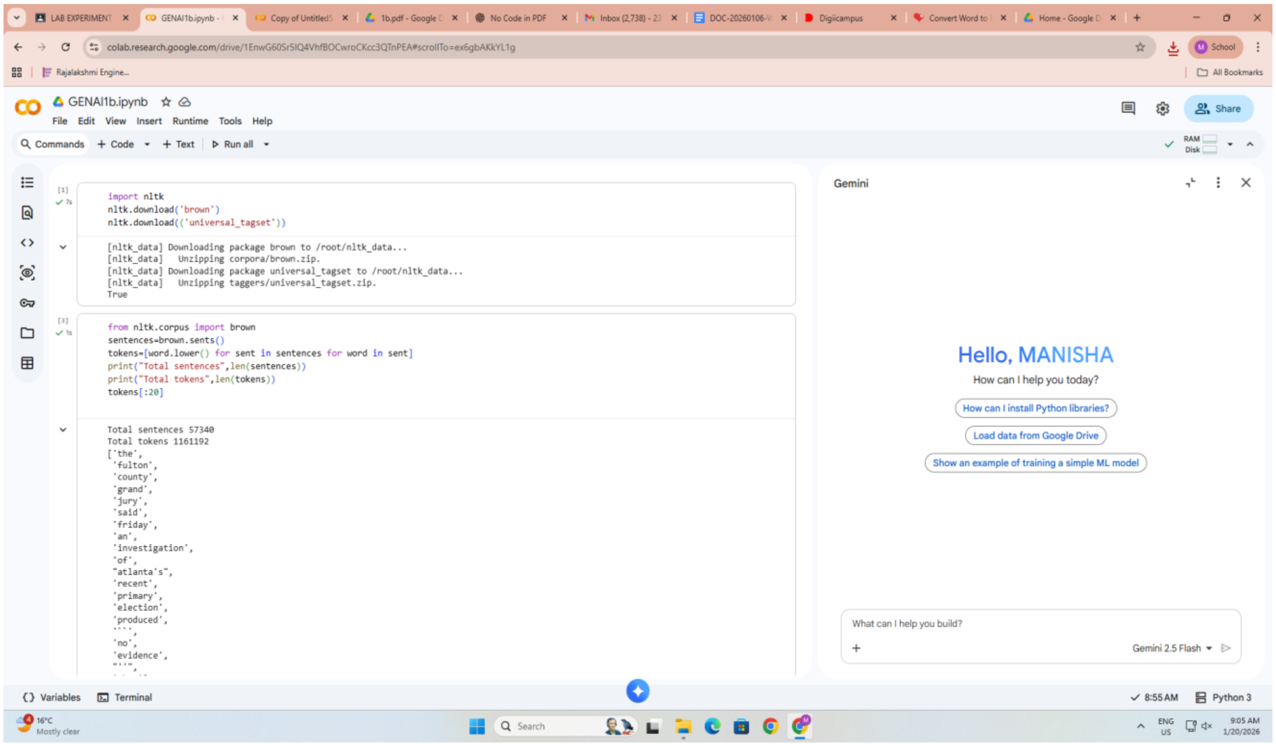
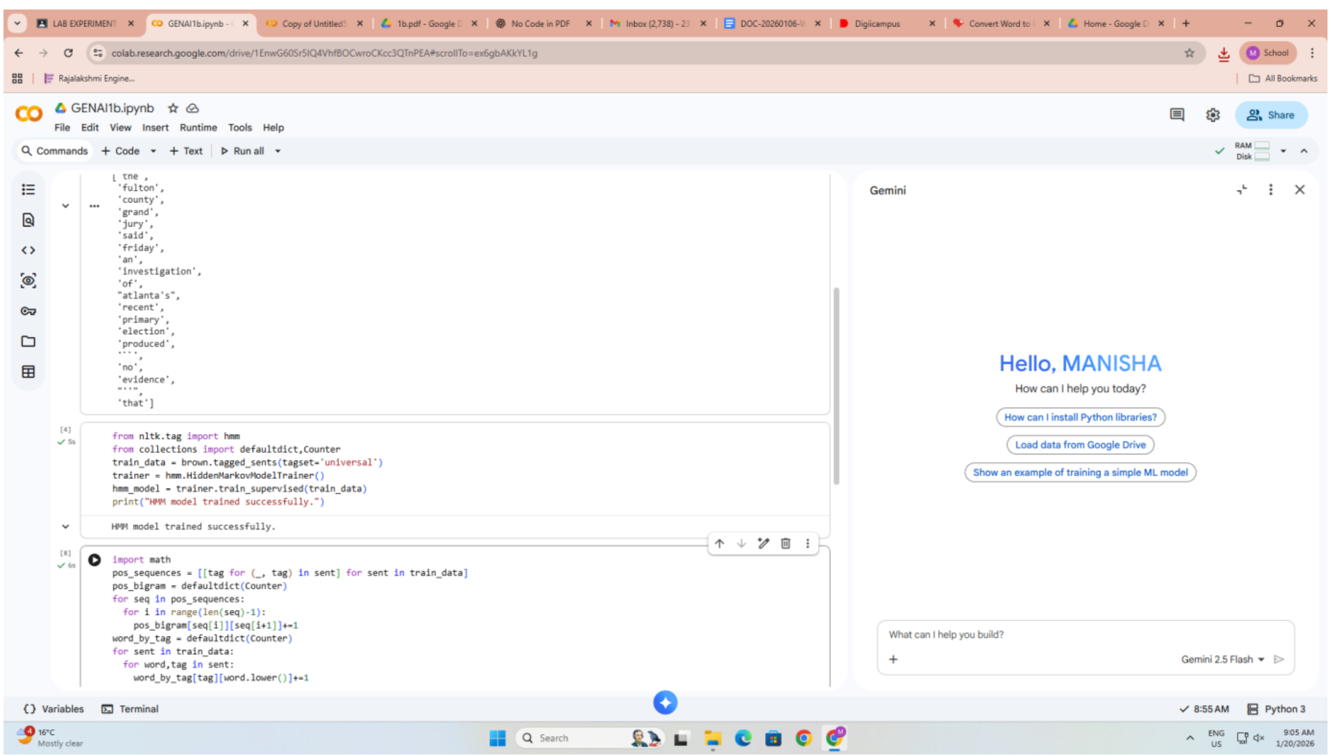
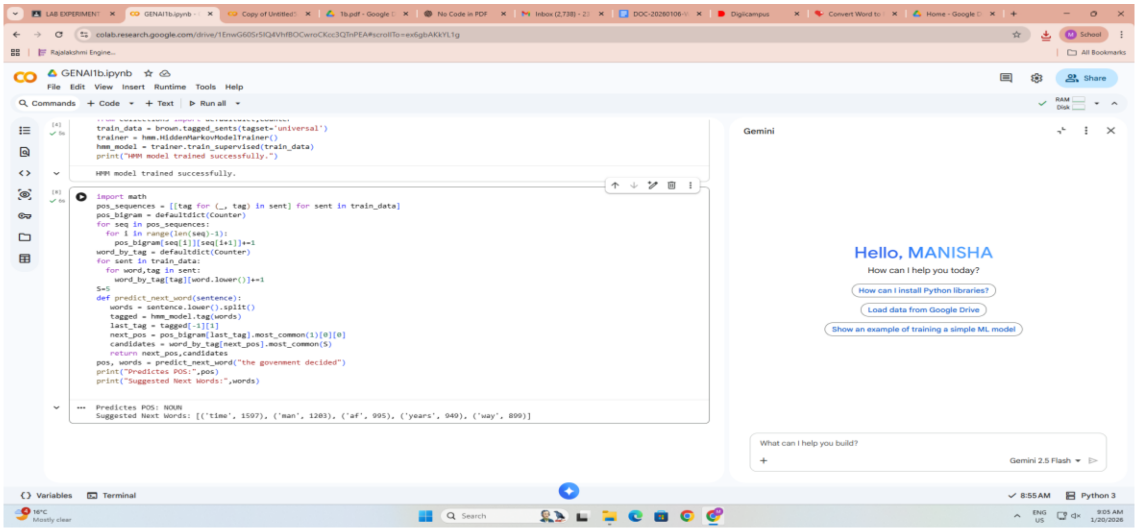
Ex1b: Hidden Markov Model (HMM) based Predictive Text System






GENAI1b.ipynb
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train_data = brown.tagged_sents(tagset='universal')
trainer = hmm.HiddenMarkovModelTrainer()
hmm_model = trainer.train_supervised(train_data)
print("HMM model trained successfully.")
HMM model trained successfully.
import math
pos_sequences = [[tag for (_, tag) in sent] for sent in train_data]
pos_bigram = defaultdict(Counter)
for seq in pos_sequences:
for i in range(len(seq)-1):
pos_bigram[seq[i]][seq[i+1]]+=1
word_by_tag = defaultdict(Counter)
for sent in train_data:
for word,tag in sent:
word_by_tag[tag][word.lower()]+=1
S=5
def predict_next_word(sentence):
words = sentence.lower().split()
tagged = hmm_model.tag(words)
last_tag = tagged[-1][1]
next_pos = pos_bigram[last_tag].most_common(1)[0][0]
candidates = word_by_tag[next_pos].most_common(5)
return next_pos,candidates
pos, words = predict_next_word("the govenment decided")
print("Predictes POS:",pos)
print("Suggested Next Words:",words)
Predictes POS: NOUN
Suggested Next Words: [('time', 1597), ('man', 1203), ('af', 995), ('years', 949), ('way', 899)]
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